VRAJ PATEL

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SUMMARY

Software Developer with a strong foundation in computer science fundamentals and hands-on experience through academic projects and industry internship work. Experienced in C++ and Python within Linux environments, with an interest in backend and systems-oriented development. Strong problem-solving ability, with experience collaborating in team environments and communicating technical ideas clearly.

TECHNICAL SKILLS

- **Languages**: C++, C, Python, x86 Assembly (basic), Java, JavaScript, MATLAB
- **Databases**: SQL, PostgreSQL
- **Backend**: RESTful API Integration
- **Web Technologies**: HTML, CSS, Bootstrap
- **Version Control**: Git
- **Systems & Networking**: Linux (Ubuntu), TCP/IP, Multithreading, basic concurrency concepts
- **Cloud & Tools**: AWS (EC2 deployment), Docker, Jira, Confluence
- **Testing**: Unit Testing (GoogleTest), UI Testing (Selenium)
- **Core Concepts**: Data Structures and Algorithms, Object-Oriented Programming (OOP), Debugging, Code Optimization, SDLC, Agile Methodology

EDUCATION

BACHELOR OF SCIENCE – COMPUTING SCIENCE **Burnaby, BC**
Simon Fraser University *June 2025*

- Concentration: **Programming Languages and Software**
- Minor: **Mathematics**

WORK EXPERIENCE

WEB DEVELOPER **Burnaby, BC**
CJSF 90.1FM *January 2024 – May 2024*

- Designed and implemented responsive web components using HTML, JavaScript, and CSS, improving UI consistency across devices, reducing layout issues by approximately 15%.
- Converted Figma mockups into responsive production web pages and maintained Drupal CMS content, supporting consistent publishing processes and accessibility standards across 10+ pages.
- Automated livestream setup by integrating Stream Deck with OBS Studio, eliminating manual broadcast steps.

TECHNICAL WRITER INTERN **Waterloo, ON**
BlackBerry (IVY and Cybersecurity) *September 2022 – April 2023*

- Worked with C++ and Python in a Linux environment to test and validate backend APIs and SDK components.
- Identified and resolved pre-release defects, implementing C++ fixes and updating validation scripts through code reviews, decreasing recurring issue reports by an estimated 20%.
- Built Python assertion-based tests to validate application logic and expected outputs, improving automated test coverage.
- Deployed a server-hosted SQL solution replacing manual Excel uploads, reducing processing time by over 30% and enabling structured data ingestion into Adobe Experience Manager (AEM) across Cylance products.
- Authored and enforced 30+ XML validation rules (Schematron), reducing publishing inconsistencies by approximately 40% and improving documentation quality.
- Reviewed Siteimprove accessibility findings, applying AEM frontend fixes to ensure WCAG compliance and improve site usability.

PROJECTS

HEART DISEASE PREDICTION MODEL (View on GitHub)
Technologies used: Python, Numpy, Pandas, Matplotlib, Scikit-Learn, Statsmodels, Jupyter Notebook

- Developed a binary classification model to predict heart disease risk using clinical and demographic indicators from a large-scale public health survey dataset.
- Prepared the dataset by cleaning records, converting categorical variables into numeric values, and removing outliers before model training.
- Trained and benchmarked multiple machine learning models (Linear Regression, Random Forest, KNN, XGBoost) using an 80/20 train-test split, with hyperparameter tuning.
- Achieved 91.4% accuracy and evaluated precision/recall trade-offs to address imbalanced data.

MULTI-CLIENT TCP CHAT SERVER (View on GitHub)
Technologies used: C, Pthreads, Linux, TCP/IP

- Built a multi-threaded TCP server in C that handles concurrent client connections and broadcasts messages.
- Used pthreads for concurrency and implemented basic synchronization to ensure correct message handling.
- Implemented a simple message protocol that includes sender IP/port information and supports client termination.
- Developed concurrent client programs to simulate network traffic and validate synchronization, thread behaviour and socket communication.

GAMEINN - Full-Stack Gaming Platform (View on GitHub)
Technologies used: HTML, CSS, JavaScript, Bootstrap, Java

- Designed a Java backend with PostgreSQL to manage user authentication, group matching logic, and database transactions.
- Developed backend logic for a group finder feature, allowing users to create and join gaming queues based on shared preferences.
- Integrated the IGDB API to fetch and display top game titles and built a searchable clip-sharing page for users to post and discover gaming content.

SCHOOL MANAGEMENT SYSTEM (View on GitHub)
Technologies used: C++

- Built a C++ desktop application using object-oriented design to manage students, courses, enrollments, and letter grades, including GPA calculation and top-student ranking.
- Engineered custom data structures (SmarterArray<T> and Map<K,V>) to manage records with dynamic resizing, memory allocation, and assertions to prevent invalid access.

TEACHING EXPERIENCE

VOLUNTEER TEACHING ASSISTANT – Python Programming **Remote**
Public Policy in Africa Initiative (PPiAI) *January 2021 – December 2022*
Published Book | ***Sample Lecture***

- Supported remote Python programming sessions alongside a lead data engineer, guiding students through coding exercises, debugging techniques, and core programming concepts.
- Contributed to preparing and organizing lecture materials later published as a textbook.
- Provided one-on-one support to students during live sessions, reinforcing problem-solving and programming fundamentals.

SALES WORK EXPERIENCE

RETAIL TEAM LEAD **Burnaby, BC**
Mountain Warehouse, Metropolis *August 2019 – Present*

- Supervised daily store operations, coordinating associates and stock handling to ensure smooth and efficient floor execution.
- Managed product levels and organized replenishment to maintain accurate and consistent stock availability.